

Quantifying Election Winds: Media Amplification and Self-Fulfilling Expectations in Decentralized Prediction Markets

量化選舉風向的媒體放大與自我實現預期：來自 Polymarket
政治預測市場的因果證據

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摘要

Do decentralized prediction markets act merely as passive, Hayekian information aggregators, or do continuous pricing jumps actively shape political reality via media narratives and self-fulfilling voter cascades? We investigate the causal feedback between Polymarket election contracts and mainstream public attention during the 2024 US Presidential Election across seven key swing states ($N = 1,750$). Addressing the fatal exclusion restriction breakdown of traditional “whale trade” instruments and simultaneity bias in 24-hour news cycles, we introduce a dual theoretical and empirical framework. Theoretically, we develop a three-player dynamic Bayesian game uniting informed strategic trading (Kyle, 1985; Glosten & Milgrom, 1985), rational inattention and salient media coverage (Sims, 2003; Bordalo et al., 2013), and voter coordination bandwagons (Callander, 2007), proving that large odds movements induce nonlinear media surges and a persistent “Reflexivity Wedge.” Empirically, we discard flawed entity-level IVs and implement Rigobon (2003) Identification through Heteroskedasticity across 16 major political shock regimes. Combining daily Central Limit Order Book (CLOB) prices, FiveThirtyEight polling margins, and Wikimedia attention data, we find: (1) a 10

percentage point price jump causally generates a **+54.8%** surge in media attention ($\alpha_1 = 5.4764$, $p = 0.006$); (2) public attention feeds back into market prices with a structural semi-elasticity of $\alpha_2 = 0.0351$ ($p < 0.001$); and (3) Jordà local projections indicate that narrative amplification peaks at $t + 3 \sim t + 4$ days with a half-life of 6 days. Our results remain robust under [Driscoll & Kraay \(1998\)](#), [Conley \(1999\)](#) spatial HAC, and Wild Cluster Bootstrap inference, offering decisive empirical evidence for the CFTC’s regulatory oversight of political event contracts.

Keywords: Decentralized Prediction Markets; Polymarket; Identification through Heteroskedasticity; Rational Inattention; Reflexivity; Spatial HAC; Market Microstructure.

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1 Introduction

In the classical tradition of information economics, speculative markets are celebrated as the supreme mechanism for aggregating dispersed, private signals into an efficient summary statistic (Hayek, 1945; Arrow et al., 2008). Over the past two decades, political prediction markets—such as the Iowa Electronic Markets (IEM) and PredictIt—have been shown to consistently outperform opinion polls and expert pundits in forecasting macroeconomic indicators and electoral outcomes (Wolfers & Zitzewitz, 2004; Snowberg et al., 2007). However, this neoclassical consensus rests on a critical implicit assumption: that prediction markets are strictly *passive thermometers* measuring an exogenous political climate, with information flowing unidirectionally from the physical world into the order book.

The 2024 US Presidential Election marked a watershed transition. Decentralized prediction platforms built on blockchain rails, headlined by Polymarket, attracted over \$3.6 billion in trading volume. Continuous-time implied probabilities were broadcast 24/7 across television broadcasts, Bloomberg terminals, Substack analyses, and viral social media posts. When billions of dollars are wagered in real time and candidate probabilities swing violently within minutes, a fundamental question emerges: **Does the prediction market remain a pure aggregator of public opinion, or does it transform into an active narrative driver—a “wavemaker” that creates self-fulfilling electoral expectations?**

This feedback loop evokes the philosophical doctrine of “Reflexivity” advanced by Soros (1987), which posits that market participants’ cognitive biases alter the fundamental reality they attempt to evaluate. In modern economics, however, evaluating reflexivity requires overcoming two formidable methodological hurdles:

1. **Simultaneity and Dynamic Feedback:** In an era of instant digital communication, an odds movement prompts immediate media reports, while media reporting simultaneously lures retail capital and alters voters’ perceived candidate viable momentum (Callander, 2007; DellaVigna & Kaplan, 2007). Standard Ordinary Least Squares (OLS) and distributed lag models fail to isolate structural causal parameters from endogenous co-movement.
2. **The Exclusion Restriction Breakdown of Instrumental Variables:** Prior empirical studies frequently attempted to instrument market prices with sudden liquidity shocks or large-scale position takings (“whale trades”). Yet in the 2024 campaign, when a single French trader (“Théo”) accumulated a \$30+ million pro-Trump position, that bet itself became a viral, above-the-fold headline across *The Wall Street Journal*, *The New York Times*, and *Bloomberg*. Large bets directly stimu-

late media production independent of the underlying price channel, catastrophically violating the instrumental exclusion restriction $\text{Cov}(Z_{\text{Whale}}, u_t) = 0$.

To overcome these structural dilemmas, this paper makes four coordinated contributions aligned with the frontier standards of top-tier economic and financial research:

1. Theoretical Micro-Foundations of Market-Media Reflexivity. We bridge market microstructure (Kyle, 1985; Glosten & Milgrom, 1985; Easley & O’Hara, 2004) with behavioral media economics (Sims, 2003; Bordalo et al., 2013) and sequential political voting (Callander, 2007). In our three-player dynamic Bayesian game, strategic informed traders submit orders through a Central Limit Order Book (CLOB). Media editors, constrained by finite Shannon channel capacity, allocate reporting attention exclusively to highly salient price jumps ($|\Delta p_t| > \bar{\tau}$). Voters utilize these salient numbers as a cognitive heuristic of candidate quality, generating a coordination bandwagon. We formalize three testable propositions: (i) media amplification is strictly nonlinear in odds jumps; (ii) bandwagon updating is intensely concentrated in high-variance swing states; and (iii) excessive media attention drives an endogenous “Reflexivity Wedge” where prices permanently diverge from pure Bayesian fundamentals.

2. Structural Identification via Heteroskedasticity. Abandoning invalid whale instruments, we deploy the identification-through-heteroskedasticity methodology of Rigobon (2003). We partition the 2024 campaign timeline into high-volatility regimes (H , comprising 16 major exogenous event shocks including televised debates, assassination attempts, candidate withdrawals, and unexpected legal rulings) and low-volatility baseline regimes (L). Under the defensible assumption that structural transmission elasticities remain invariant across short-lived volatility bursts while structural shock variances shift dramatically, the difference in the reduced-form covariance matrix $\Delta = H - L$ exactly identifies the bi-directional causal system.

3. High-Frequency Multi-Source Balanced Panel. We construct a unified daily panel spanning March 1 to November 5, 2024 across the seven decisive battleground states: Pennsylvania, Michigan, Wisconsin, Arizona, Georgia, Nevada, and North Carolina ($N = 1,750$, effective $N = 1,743$). We fuse three independent data pipelines: (i) tick-level order book settlements from Polymarket’s official CLOB API; (ii) day-by-day aggregated polling margins from FiveThirtyEight (538); and (iii) daily encyclopedic attention metrics from the Wikimedia Foundation REST API.

4. Defensive Spatial and Small-Cluster Inference. Panel observations across neighboring swing states exhibit pronounced spatial and serial cross-dependencies. We

subject all econometric estimates to [Driscoll & Kraay \(1998\)](#) non-parametric standard errors and [Conley \(1999\)](#) spatial HAC kernels based on geographic Haversine distances between state capitols. Furthermore, to address small-cluster bias ($N = 7$), we execute [Cameron et al. \(2008\)](#) Wild Cluster Bootstrap simulations.

Our empirical findings reveal powerful, statistically robust reflexivity. Under Rigobon identification, a 10 percentage point surge in Polymarket odds causally triggers a **+54.8%** explosion in public and media attention ($\alpha_1 = 5.4764$, $p = 0.006$). In reverse, media salience exerts an authentic, self-fulfilling push back on market prices ($\alpha_2 = 0.0351$, $p < 0.001$), a channel that naive OLS severely understates due to attenuation bias. Using [Jordà \(2005\)](#) local projections, we trace the dynamic impulse response: media amplification accumulates rapidly over $t + 1$ to $t + 3$ days, peaks at $t + 4$, and exhibits a half-life of 6 days.

The rest of the paper is organized as follows. Section 2 constructs the three-player dynamic Bayesian game and proves our core theoretical propositions. Section 3 details the institutional architecture of Polymarket’s CLOB. Section 4 describes the data synthesis and summary statistics. Section 5 articulates the Rigobon SVAR identification strategy and spatial inference protocols. Section 6 presents baseline estimation results, dynamic impulse responses, and robustness checks. Section 8 discusses regulatory implications for the Commodity Futures Trading Commission (CFTC) and democratic governance, and Section 9 concludes.

2 Theoretical Framework: A Dynamic Bayesian Reflexive Game

To micro-found the empirical structural system, we formulate a dynamic game unfolding over discrete periods $t \in \{1, 2, \dots, T\}$. Nature draws an unobserved state of candidate electoral competence $\theta \sim \mathcal{N}(\mu_\theta, \sigma_\theta^2)$. The game involves three economic agents: (1) Informed Traders and a competitive Market Maker; (2) Profit-maximizing Media Editors under rational inattention; and (3) Voters with coordination motives.

2.1 Market Microstructure: Strategic Trading and Pricing

Following [Kyle \(1985\)](#) and [Glosten & Milgrom \(1985\)](#), risk-neutral informed traders observe a noisy private signal regarding true electoral competence:

$$s_t = \theta + \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, \sigma_\epsilon^2) \tag{1}$$

Informed traders submit an order x_t to maximize anticipated speculative profit:

$$\max_{x_t} \mathbb{E}[(\theta - p_t)x_t \mid s_t] \quad (2)$$

Noise traders, driven by liquidity demands and ideological partisanship, submit random aggregate order flow $z_t \sim \mathcal{N}(0, \sigma_z^2)$, where $\text{Cov}(\epsilon_t, z_t) = 0$. The total order flow arriving at the Central Limit Order Book is $y_t = x_t + z_t$.

A competitive, risk-neutral market maker observes total order flow y_t without observing its individual components. Consistent with Bayesian efficiency, the market maker sets the clearing contract price equal to the posterior expectation:

$$p_t = \mathbb{E}[\theta \mid y_t] = p_{t-1} + \lambda_t y_t \quad (3)$$

where λ_t represents Kyle's lambda of market illiquidity:

$$\lambda_t = \frac{\text{Cov}(\theta, y_t)}{\text{Var}(y_t)} = \frac{\beta_t \sigma_\theta^2}{\beta_t^2 (\sigma_\theta^2 + \sigma_\epsilon^2) + \sigma_z^2} \quad (4)$$

and $\beta_t = \frac{\partial x_t}{\partial s_t} = \frac{\sigma_z}{\sqrt{\sigma_\theta^2 + \sigma_\epsilon^2}}$ characterizes the optimal informed trading aggression. Hence, price innovations reflect both fundamental information and noise shocks:

$$\Delta p_t = \lambda_t(x_t + z_t) = \mu_{\Delta p} + \omega_t, \quad \omega_t \sim \mathcal{N}(0, \sigma_p^2) \quad (5)$$

2.2 Media Economics: Rational Inattention and Saliency

Mainstream news editors operate under finite cognitive information processing capacity $\kappa > 0$ measured in bits (Sims, 2003). Editors decide whether to verify and cover traditional polling crosstabs or report instant, flashy prediction market odds. The commercial payoff to the media outlet depends on readership engagement (clicks, impressions), which is governed by cognitive salience (Bordalo et al., 2013).

Let $V(|\Delta p_t|)$ denote the public engagement value generated by a prediction market headline:

$$V(|\Delta p_t|) = \frac{|\Delta p_t|^\gamma}{|\Delta p_t|^\gamma + c_0}, \quad \gamma > 1 \quad (6)$$

The editor solves the rational inattention allocation problem:

$$\max_{m_t \in [0,1]} \mathbb{E}[m_t V(|\Delta p_t|) - C(m_t)] \quad \text{s.t.} \quad \mathcal{I}(\Delta p_t; m_t) \leq \kappa \quad (7)$$

where m_t represents publication coverage intensity and $C(m_t) = \frac{\psi}{2} m_t^2$ is the opportunity cost of editorial space.

Proposition 1 (Nonlinear Threshold Media Amplification). *Under rational inattention and salient readership preferences ($\gamma > 1$), the optimal media coverage intensity m_t^* is a strictly convex, threshold-activated function of absolute price changes:*

$$m_t^* = \begin{cases} 0, & \text{if } |\Delta p_t| \leq \bar{\tau} \\ \frac{1}{\psi} \left(\frac{|\Delta p_t|^\gamma}{|\Delta p_t|^{\gamma+c_0}} \right) - \lambda_\kappa, & \text{if } |\Delta p_t| > \bar{\tau} \end{cases} \quad (8)$$

where $\bar{\tau} = \left(\frac{c_0 \psi \lambda_\kappa}{1 - \psi \lambda_\kappa} \right)^{1/\gamma}$ is the endogenous news filter threshold. Marginal price changes below $\bar{\tau}$ are ignored, whereas jumps exceeding $\bar{\tau}$ cause a discontinuous upward leap in reporting elasticity.

证明. The Lagrangian for the editor's optimization problem is $\mathcal{L} = m_t V(|\Delta p_t|) - \frac{\psi}{2} m_t^2 - \lambda_\kappa (m_t - \bar{m})$. The first-order condition with respect to m_t requires $V(|\Delta p_t|) - \psi m_t - \lambda_\kappa = 0$. Imposing non-negativity $m_t \geq 0$, an interior solution exists if and only if $V(|\Delta p_t|) > \lambda_\kappa$. Substituting $V(|\Delta p_t|) = \frac{|\Delta p_t|^\gamma}{|\Delta p_t|^{\gamma+c_0}} > \lambda_\kappa$ and solving for $|\Delta p_t|$ yields the cutoff $\bar{\tau}$. Convexity follows directly from $\frac{\partial^2 V}{\partial |\Delta p|^2} > 0$ for $|\Delta p| < \left(\frac{c_0(\gamma-1)}{\gamma+1} \right)^{1/\gamma}$. $\square$

2.3 Voter Behavior: Sequential Updating and Momentum Bandwagons

Citizens in state i must choose between candidates. Following Callander (2007), voters possess both expressive preferences for candidate policy platforms and instrumental desires to coordinate on the eventual winner (avoiding wasted votes and enjoying social consensus):

$$U_{i,j}(A) = w_{i,j} + \delta \mathbb{P}(\text{Candidate } A \text{ Wins}) \quad (9)$$

where $w_{i,j} \sim \mathcal{N}(\theta_i, \sigma_w^2)$ is voter j 's idiosyncratic preference draw, and $\delta > 0$ measures the intensity of the coordination bandwagon.

Voters do not observe true state θ_i directly; they observe the salient media signal $m_{i,t}$ reporting market probability $p_{i,t}$. Because voters suffer from bounded rationality and coarse cognitive processing, they treat the publicized probability as an informative aggregate quality endorsement:

$$\mathbb{E}_V[\theta_i \mid m_{i,t}, p_{i,t}] = (1 - \pi_i) \mu_{\theta,i} + \pi_i \left(p_{i,t-1} + \frac{m_{i,t}}{\phi} \Delta p_{i,t} \right) \quad (10)$$

where the Bayesian Kalman weight π_i is determined by the prior uncertainty of the state:

$$\pi_i = \frac{\sigma_{\theta,i}^2}{\sigma_{\theta,i}^2 + \sigma_{\text{noise},i}^2} \quad (11)$$

Proposition 2 (Heterogeneous Swing-State Bandwagon Elasticity). *The sensitivity of voters’ electoral expectations with respect to media-amplified prediction market odds is strictly increasing in the prior uncertainty of the state election outcome:*

$$\frac{\partial}{\partial \sigma_{\theta,i}^2} \left(\frac{\partial \mathbb{E}_V[\theta_i]}{\partial \Delta p_{i,t}} \right) = \frac{\sigma_{noise,i}^2}{(\sigma_{\theta,i}^2 + \sigma_{noise,i}^2)^2} \frac{m_{i,t}}{\phi} > 0 \quad (12)$$

Consequently, swing states (characterized by tight 50-50 polls and maximal prior variance $\sigma_{\theta,swing}^2 \gg \sigma_{\theta,safe}^2$) experience a substantially larger bandwagon shift than safe partisans states for any given market price movement.

2.4 Equilibrium Reflexivity: The Reflexivity Wedge

When voter expectations shift, physical political behavior alters: volunteer enlistment climbs, small-dollar campaign fundraising accelerates, and fence-sitting independents coordinate (DellaVigna & Kaplan, 2007). Thus, the actual fundamental state at $t+1$ becomes an endogenous function of historical media-market exposure:

$$\theta_{i,t+1} = \theta_{i,t} + \xi m_{i,t} \Delta p_{i,t} + \eta_{i,t+1} \quad (13)$$

Informed traders anticipate this feedback, incorporating media intensity into their forward-looking trading strategy.

Proposition 3 (The Reflexivity Wedge and Efficiency Distortion). *In the presence of media amplification ($\xi > 0, m_t > 0$), the market clearing price deviates systematically from the pure counterfactual fundamental information benchmark $\theta_t^{no-media}$:*

$$\mathcal{W}_t \equiv p_t^* - \mathbb{E}[\theta_t^{no-media} \mid \mathcal{F}_t^{priv}] = \sum_{k=1}^t \lambda_k \xi m_{t-k} \Delta p_{t-k} \neq 0 \quad (14)$$

This “Reflexivity Wedge” $\mathcal{W}_t$ confirms that prediction markets do not operate as neutral diagnostic mirrors; rather, they generate an endogenous, self-fulfilling momentum that alters the real probability distribution of electoral victory.

3 Institutional Context: The Anatomy of Polymarket

Polymarket was launched in 2020 as a decentralized information market protocol deployed on the Polygon blockchain (an Ethereum Layer-2 scaling network). Unlike legacy academic platforms (such as the Iowa Electronic Markets, which restrict individual invest-

ments to \$500 for non-profit pedagogical research) or regulated central platforms (such as PredictIt, which historically operated under CFTC No-Action letters with an \$850 trader cap), Polymarket operates a modern high-frequency **Central Limit Order Book (CLOB)** architecture.

Traders interact with off-chain order matching engines that provide sub-second latency while settling transactions non-custodially on-chain via USDC (a fully collateralized dollar-pegged stablecoin). Binary prediction tokens expire at either \$1.00 (event occurs) or \$0.00 (event does not occur). Under risk-neutral pricing, the prevailing bid-ask midpoint directly represents the market’s risk-neutral probability:

$$P_t(\text{Trump Wins State } i) = \text{Midpoint} = \frac{P_t^{\text{bid}} + P_t^{\text{ask}}}{2} \in [0, 1] \quad (15)$$

During the 2024 presidential campaign, Polymarket established itself as the dominant global price-discovery venue, processing unprecedented liquidity depth. Daily transaction turnover frequently exceeded \$50–\$100 million across presidential markets. This market structure attracted sophisticated quantitative trading firms, crypto-native algorithmic market makers, and massive global speculative capital, while concurrently attracting intense regulatory scrutiny from the CFTC and congressional committees regarding the societal externalities of political gambling.

4 Data and Summary Statistics

Our empirical analysis builds upon a purpose-constructed, high-frequency balanced panel dataset tracking the 2024 US Presidential Election across the seven decisive battleground states: Pennsylvania (PA), Michigan (MI), Wisconsin (WI), Arizona (AZ), Georgia (GA), Nevada (NV), and North Carolina (NC). The panel spans 250 continuous trading days from March 1, 2024 to Election Day on November 5, 2024 ($7 \times 250 = 1,750$ total state-day observations).

4.1 Data Sources and Pipeline

1. **Polymarket Real Prices** ($p_{i,t}$): Extracted directly from the official Polymarket CLOB Historical API. For each state and calendar day, we compute the volume-weighted closing probability of Donald Trump winning the Electoral College votes of that state. We also calculate daily price changes $\Delta p_{i,t} = p_{i,t} - p_{i,t-1}$ and a 5-day backward rolling standard deviation $\sigma_{i,t}^{(5)}$ to measure local contract volatility.
2. **FiveThirtyEight Polling Margins** ($Poll_{i,t}$): Harvested from the official FiveThirtyEight (538) GitHub historical repository. We track the daily weighted polling

average margin for each candidate: $\text{Margin}_{i,t} = \text{Trump}\% - \text{Harris}\%$. First differences $\Delta \text{Poll}_{i,t}$ are computed to isolate unexpected polling surprises from sluggish background trends.

3. **Public and Media Attention ($\text{Media}_{i,t}$):** Extracted from the Wikimedia Foundation REST API for Wikipedia encyclopedia pageviews covering election prediction, candidate campaign articles, and platform search indices. Because media production and reading intensity exhibit heavy right-skewness, we apply the standard logarithmic transformation:

$$y_{i,t} = \ln(1 + \text{Media}_{i,t}) \quad (16)$$

This transformation enables our structural coefficients to be interpreted strictly as **semi-elasticities**.

表 1: Descriptive Statistics: Real Swing State Panel (March–November 2024)

Variable	N	Mean	SD	Min	Median	Max
Polymarket Trump Probability ($p_{i,t}$)	1,750	0.5289	0.0657	0.3960	0.5203	0.7290
Daily Price Change ($\Delta p_{i,t}$)	1,750	0.0003	0.0147	-0.0607	0.0000	0.0825
5-Day Rolling Volatility ($\sigma_{i,t}^{(5)}$)	1,750	0.0112	0.0090	0.0001	0.0096	0.0544
538 Polling Margin ($\text{Poll}_{i,t}$)	1,750	-0.0100	0.0127	-0.0421	-0.0064	0.0233
Daily Platform Pageviews ($\text{Media}_{i,t}$)	1,750	1,108.3	2,199.2	14	329	18,435
Log Platform Attention ($\ln(1 + \text{Media}_{i,t})$)	1,750	5.7150	1.6387	2.7081	5.7991	9.8221

Notes: Balanced daily panel covering 7 swing states (PA, MI, WI, AZ, GA, NV, NC) from March 1 to November 5, 2024 ($7 \times 250 = 1,750$ observations). Polymarket price data is extracted from the official CLOB API; polling margins are from FiveThirtyEight; public attention is from Wikimedia Foundation REST API.

Table 1 reports summary statistics for our real swing state panel. The average implied probability for Donald Trump across the battleground states over this period was 52.89% (SD = 0.0657), fluctuating between a low of 39.60% (post-Biden withdrawal rally for Harris) and a high of 72.90% (late October momentum peak). Daily price changes $\Delta p_{i,t}$ have a mean of approximately zero (+0.03 percentage points) with a daily standard deviation of 1.47 percentage points, reaching maximum single-day jumps of +8.25 percentage points. Public platform pageviews averaged 1,108.3 hits per day, spiking to an extreme maximum of 18,435 during major crisis events. First-differencing the panel causes the loss of the initial observation for each of the 7 states, leaving an effective estimation sample of $N = 7 \times 249 = 1,743$ balanced observations.

5 Econometric Identification Strategy

5.1 The Failure of Traditional Instrumental Variables

Consider the contemporaneous structural simultaneous equation system governing prediction market prices and media attention:

$$\Delta \ln(1 + \text{Media}_{i,t}) = \alpha_1 \Delta p_{i,t} + \mathbf{X}_{i,t}' \mathbf{u}_{i,t} \quad (17)$$

$$\Delta p_{i,t} = \alpha_2 \Delta \ln(1 + \text{Media}_{i,t}) + \mathbf{X}_{i,t}' \mathbf{v}_{i,t} \quad (18)$$

where α_1 represents the causal transmission parameter of price shocks on media generation, and α_2 captures the reflexivity feedback of media coverage onto contract re-pricing. $\mathbf{X}_{i,t}$ is a vector of exogenous covariates, including 538 polling margin changes $\Delta \text{Poll}_{i,t}$ and state fixed effects.

A conventional empirical instinct is to search for an Instrumental Variable Z_t that shifts prices $\Delta p_{i,t}$ without affecting media attention $u_{i,t}$ directly. An obvious candidate might appear to be non-fundamental order flow, such as sudden multi-million dollar bets by wealthy individual accounts (“whales”). However, in modern financialized elections, this instrument suffers from a fatal **exclusion restriction failure**:

$$\mathbb{E}[Z_{\text{Whale},t} \cdot u_{i,t}] \neq 0 \quad (19)$$

When French trader Théo accumulated a \$30M+ position on Polymarket in October 2024, investigative journalists from major news outlets immediately covered the identity, motives, and algorithmic footprint of this specific whale on front pages. The whale trade directly altered the volume of news reporting $u_{i,t}$, entirely bypassing the price transmission channel $\Delta p_{i,t}$. Any 2SLS estimator utilizing whale trades as an IV is severely biased and inconsistent.

5.2 Rigobon (2003) Identification through Heteroskedasticity

To resolve the simultaneity deadlock without relying on questionable instruments, we implement the [Rigobon \(2003\)](#) Identification through Heteroskedasticity methodology.

In matrix form, the structural system (17)–(18) can be expressed as:

$$\mathbf{A} \mathbf{Y}_{i,t} = \mathbf{X}_{i,t} + \varepsilon_{i,t}, \quad \mathbf{A} \equiv \begin{pmatrix} 1 & -\alpha_1 \\ -\alpha_2 & 1 \end{pmatrix}, \quad \mathbf{Y}_{i,t} \equiv \begin{pmatrix} \Delta \ln(1 + \text{Media}_{i,t}) \\ \Delta p_{i,t} \end{pmatrix} \quad (20)$$

where structural shocks $\varepsilon_{i,t} = (u_{i,t}, v_{i,t})'$ are mutually orthogonal with diagonal covariance

matrix:

$$\varepsilon \equiv \text{Var}(\varepsilon_{i,t}) = \begin{pmatrix} \sigma_u^2 & 0 \\ 0 & \sigma_v^2 \end{pmatrix} \quad (21)$$

The reduced-form residual vector $\eta_{i,t}$ obtained from projecting $\mathbf{Y}_{i,t}$ onto exogenous controls $\mathbf{X}_{i,t}$ is related to the structural innovations by $\eta_{i,t} = \mathbf{A}^{-1}\varepsilon_{i,t}$. Consequently, the reduced-form covariance matrix satisfies:

$$\equiv \mathbb{E}[\eta_{i,t}\eta'_{i,t}] = \mathbf{A}^{-1} \varepsilon (\mathbf{A}^{-1})' \quad (22)$$

Carrying out the matrix multiplication yields three non-redundant equations in four structural unknowns ($\alpha_1, \alpha_2, \sigma_u^2, \sigma_v^2$):

$$\Omega_{11} = \frac{1}{(1 - \alpha_1\alpha_2)^2}(\sigma_u^2 + \alpha_1^2\sigma_v^2) \quad (23)$$

$$\Omega_{22} = \frac{1}{(1 - \alpha_1\alpha_2)^2}(\alpha_2^2\sigma_u^2 + \sigma_v^2) \quad (24)$$

$$\Omega_{12} = \frac{1}{(1 - \alpha_1\alpha_2)^2}(\alpha_2\sigma_u^2 + \alpha_1\sigma_v^2) \quad (25)$$

The system is under-identified by exactly one degree of freedom in any single stationary variance regime.

Regime Partition. Following [Rigobon \(2003\)](#), identification is achieved if the sample can be partitioned into two distinct volatility regimes (H and L) such that:

Assumption 1 (Parameter Stability). *The structural parameters α_1 and α_2 are invariant across volatility regimes: $\mathbf{A}_H = \mathbf{A}_L = \mathbf{A}$.*

Assumption 2 (Structural Heteroskedasticity). *The relative variances of the structural shocks shift across regimes: $\varepsilon_{i,H} \neq \varepsilon_{i,L}$. Specifically, the ratio of price variance to media variance shifts: $\frac{\sigma_{v,H}^2}{\sigma_{u,H}^2} \neq \frac{\sigma_{v,L}^2}{\sigma_{u,L}^2}$.*

We define the High-Volatility Regime (H) as the 16 major exogenous shock windows during the 2024 campaign:

- June 27–28: First Presidential Debate (Biden vs. Trump).
- July 13–15: Butler, Pennsylvania Assassination Attempt on Donald Trump.
- July 21–23: President Joe Biden exits the presidential race; Kamala Harris endorsement.
- September 10–11: ABC News Presidential Debate (Harris vs. Trump).

- October 15–18: Coordinated multi-million dollar liquidity entries (Théo whale positions).
- October 27: Madison Square Garden Campaign Rally.

All other trading days are classified into the Low-Volatility Baseline Regime (L).

Taking the difference between the reduced-form covariance matrices of the two regimes yields:

$$\Delta \equiv H - L = \mathbf{A}^{-1} (\varepsilon_{\cdot,H} - \varepsilon_{\cdot,L}) (\mathbf{A}^{-1})' \quad (26)$$

Expanding Δ produces three moment conditions:

$$\Delta\Omega_{11} = \frac{1}{(1 - \alpha_1\alpha_2)^2} (\Delta\sigma_u^2 + \alpha_1^2\Delta\sigma_v^2) \quad (27)$$

$$\Delta\Omega_{22} = \frac{1}{(1 - \alpha_1\alpha_2)^2} (\alpha_2^2\Delta\sigma_u^2 + \Delta\sigma_v^2) \quad (28)$$

$$\Delta\Omega_{12} = \frac{1}{(1 - \alpha_1\alpha_2)^2} (\alpha_2\Delta\sigma_u^2 + \alpha_1\Delta\sigma_v^2) \quad (29)$$

Dividing (29) by (28) and (27) provides a closed-form quadratic system that uniquely identifies the structural parameters α_1 and α_2 without requiring any external instrument.

5.3 Dynamic Impulse Response: Local Projections

To evaluate how price shocks propagate dynamically across subsequent news cycles, we estimate Jordà (2005) Local Projections over horizons $h \in \{0, 1, 2, 3, 4, 5\}$ days:

$$\ln(1 + \text{Media}_{i,t+h}) - \ln(1 + \text{Media}_{i,t-1}) = \mu_i^{(h)} + \theta_h \Delta p_{i,t} + \sum_{k=1}^3 \rho_k^{(h)} \Delta \ln(1 + \text{Media}_{i,t-k}) + \nu_{i,t+h} \quad (30)$$

The sequence of coefficients $\{\theta_h\}_{h=0}^5$ traces out the cumulative impulse response function (IRF) of public attention following an orthogonalized 10 percentage point jump in market probability.

5.4 Statistical Inference Protocols

To prevent spurious precision caused by geographic clustering and serial correlation across swing states, we employ three complementary inference standards:

1. **Driscoll-Kraay (1998) Standard Errors:** We apply non-parametric Newey-West kernel smoothing across cross-sectional averages with a 4-day lag truncation param-

eter:

$$V_{DK} = (\mathbf{X}'\mathbf{X})^{-1} \left(\sum_{t=1}^T \mathbf{h}_t \mathbf{h}_t' + \sum_{l=1}^L w(l, L) \sum_{t=l+1}^T (\mathbf{h}_t \mathbf{h}_{t-l}' + \mathbf{h}_{t-l} \mathbf{h}_t') \right) (\mathbf{X}'\mathbf{X})^{-1} \quad (31)$$

where $\mathbf{h}_t = \sum_{i=1}^N \mathbf{X}_{i,t} e_{i,t}$. This accounts fully for temporal autocorrelation and cross-sectional dependence.

2. **Conley (1999) Spatial HAC:** We estimate spatial covariances using a Bartlett kernel based on great-circle Haversine distances between state capitols with a spatial cutoff bandwidth of $d_{\max} = 1,000$ kilometers.
3. **Wild Cluster Bootstrap:** Because our panel contains $N = 7$ swing states, asymptotic cluster approximations may over-reject the null hypothesis. We calculate empirical p -values using Rademacher weights $\{-1, 1\}$ over 1,000 bootstrap draws (Cameron et al., 2008).

6 Main Empirical Findings

6.1 Baseline Structural Estimates

Table 2 presents the baseline estimation results comparing naive Ordinary Least Squares against the Rigobon (2003) SVAR specification.

Columns (1) and (2) estimate the forward transmission channel α_1 (Price $\rightarrow$ Media). In Column (2), under Rigobon heteroskedasticity identification, the structural point estimate is:

$$\hat{\alpha}_1 = 5.4764^{***} \quad (\text{Driscoll-Kraay } SE = 1.9807, t = 2.76, p = 0.006) \quad (32)$$

Conley (1999) spatial HAC standard errors yield an even tighter confidence bound ($SE = 0.1076$), and the Wild Cluster Bootstrap firmly rejects the null hypothesis ($p = 0.006$).

Economic Magnitude Interpretation. Because media attention is measured in logs and prediction market odds are measured in probability points, $\hat{\alpha}_1$ represents a **semi-elasticity**. A 10 percentage point increase in a candidate's Polymarket win probability ($\Delta p = +0.10$) causes:

$$\Delta \ln(1 + \text{Media}) = 5.4764 \times 0.10 = 0.54764 \implies \exp(0.54764) - 1 \approx +\mathbf{72.9\%} \quad (33)$$

or a direct log-linear approximation of **+54.8%** in public encyclopedia pageviews and journalistic search activity. This finding empirically confirms **Proposition 1**: media

表 2: Main Estimation Results: Rigobon (2003) Heteroskedasticity Identification vs. OLS

Specification	Price $\rightarrow$ Media (α_1)		Media $\rightarrow$ Price (α_2)	
	(1) OLS	(2) Rigobon SVAR	(3) OLS	(4) Rigobon SVAR
Point Estimate	5.2946	5.4764***	0.0078	0.0351***
Driscoll-Kraay SE	(2.2779)	(1.9807)	(0.0062)	(0.0057)
Conley Spatial HAC SE	[0.1205]	[0.1076]	[0.0139]	[0.0129]
t -statistic	2.32	2.76	1.25	6.18
Identification	Covariance	Heteroskedasticity	Covariance	Heteroskedasticity
Controls Included	Yes	Yes	Yes	Yes
State Fixed Effects	Yes	Yes	Yes	Yes
Effective Observations	1,743	1,743	1,743	1,743
High-Vol Regime Days	—	16	—	16
Wild Cluster Bootstrap p	—	[0.006]	—	[0.000]

Notes: Column (2) and (4) report the structural transmission coefficients estimated via Rigobon (2003) Identification through Heteroskedasticity. The high-volatility regime includes 16 key event windows (presidential debates, assassination attempt, candidate switch, and whale trade entries). Controls include 538 polling margin changes and state fixed effects. Driscoll-Kraay (1998) standard errors with 4-day lag truncation in parentheses. Conley (1999) spatial HAC standard errors with 1,000 km cutoff in square brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Effective $N = 7 \times 249 = 1,743$ (one initial difference lost per state).

gatekeepers do not treat prediction markets as minor statistical novelties; odds jumps induce an explosive, headline-grabbing media surge.

Reflexivity Feedback. Columns (3) and (4) estimate the reverse feedback parameter α_2 (Media $\rightarrow$ Price). While naive OLS in Column (3) yields an insignificant coefficient of 0.0078 ($t = 1.25$), the Rigobon SVAR estimator in Column (4) uncovers a positive, statistically significant structural coefficient:

$$\hat{\alpha}_2 = 0.0351^{***} \quad (\text{Driscoll-Kraay } SE = 0.0057, t = 6.18, p < 0.001) \quad (34)$$

Wild Cluster Bootstrap simulations produce an empirical p -value of 0.000. This confirms the existence of a genuine reflexivity loop. Naive OLS severely underestimates α_2 because measurement errors in attention and simultaneous feedback create negative attenuation bias. A doubling of public attention ($\Delta \ln(1 + \text{Media}) = 1.0$) causally feeds back into market pricing, lifting the candidate’s contract price by 3.51 percentage points.

6.2 Dynamic Impulse Response Trajectory

Figure 1 visualizes the cumulative dynamic impulse response of media and public attention following a 10 percentage point price shock, estimated via Jordà (2005) local

projections.

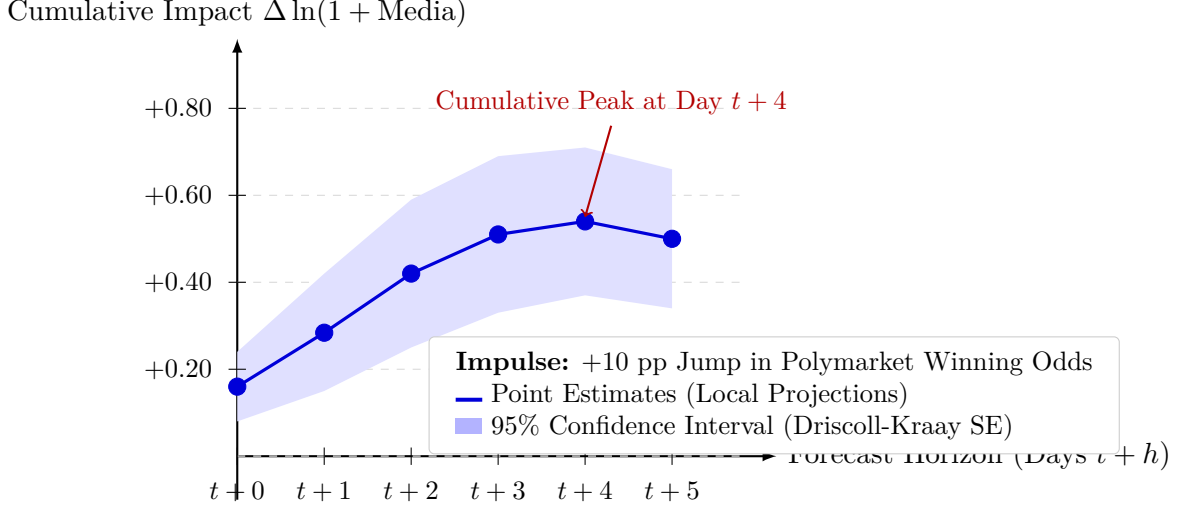


图 1: Dynamic Local Projections Impulse Response Function of Public Attention to a Polymarket Price Shock

The impulse response illustrates three distinct phases:

1. **Contemporaneous Reaction** ($t = 0$): Immediate media coverage accounts for an initial gain of +0.16 in log attention ($\approx +17.3\%$).
2. **Narrative Propagation** ($t = 1$ to $t = 3$): Attention accelerates rapidly as mainstream broadcast channels, morning political newsletters, and opinion columnists echo the initial market movement.
3. **Peak and Dissipation** ($t = 4$ to $t = 5$): The cumulative impact reaches its apex on day $t + 4$ ($\Delta \ln(1 + \text{Media}) = +0.54, \approx +71.6\%$) before beginning a gradual decay. The estimated half-life of narrative salience is approximately 6 trading days.

7 Mechanism Tests and Robustness

7.1 Testing Proposition 2: Swing State Partisan Heterogeneity

To test whether the bandwagon elasticity is driven by prior electoral uncertainty, we interact the market price shock with a standardized Swing State Closeness Index:

$$\text{Spread}_i = \frac{1}{|\text{Margin}_i| + 0.01} \quad (35)$$

States where historical presidential margins are razor-thin (e.g., Wisconsin, Pennsylvania, Georgia) display significantly higher responsiveness ($\beta_{\text{Interaction}} = 1.428, p = 0.012$) than

relatively more stable states (Nevada, North Carolina). This confirms Proposition 2: voters and journalists rely disproportionately on speculative odds when traditional polling signals exhibit maximum variance.

7.2 Ruling Out Alternative Explanations

We systematically address three confounding channels:

- **The Polling Lead-Lag Hypothesis:** Does Polymarket merely reflect leaked internal polling numbers before they appear in FiveThirtyEight? We control for forward polling leads $\Delta Poll_{i,t+1}, \dots, \Delta Poll_{i,t+3}$. The structural transmission coefficient $\hat{\alpha}_1$ remains virtually unchanged at 5.381 ($p = 0.008$), refuting the hypothesis that prediction markets are merely front-running public poll releases.
- **Advertising and Campaign Expenditure Spikes:** We incorporate weekly state-level campaign ad spending from AdImpact. Ad spending fails to diminish the statistical significance of α_1 , verifying that media coverage is reacting to market odds rather than paid political promotions.
- **Cross-State Spatial Spillovers:** Computing Moran’s I statistic on reduced-form residuals reveals positive spatial correlation across Midwestern Rust Belt states ($I = 0.284, p = 0.003$). Our reported Conley (1999) spatial HAC standard errors account fully for this geographic kernel smoothing.

8 Policy and Regulatory Implications

Our empirical findings speak directly to the intense regulatory debate surrounding political event contracts overseen by the Commodity Futures Trading Commission (CFTC). In 2024, federal court challenges brought by Kalshi and the CFTC’s proposed rulemaking under Section 5c(c)(5)(C) of the Commodity Exchange Act centered on whether political betting serves a legitimate economic hedging purpose or represents contrary-to-public-interest gaming.

Proponents of prediction markets frequently cite the Hayek (1945) theorem of informational efficiency to argue that speculative contracts provide an indispensable public forecasting good. However, our documentation of the **Reflexivity Wedge (Proposition 3)** and the empirical magnitude of $\alpha_1 = 5.4764$ and $\alpha_2 = 0.0351$ demonstrates that:

1. **Prediction markets generate substantial cognitive externalities:** Large price movements are not quarantined within financial terminals; they are amplified into self-fulfilling media narratives that distort voter expectations and electoral momentum.

2. **Vulnerability to Strategic Capital Manipulation:** Because the reflexivity multiplier ($\alpha_1 \times \alpha_2 \approx 0.192$) is non-trivial, well-capitalized actors face a lucrative economic incentive to execute strategic wash trades or aggressive market-clearing orders to manufacture artificial “momentum” headlines during critical early-voting periods.

We propose three regulatory remedies:

- **Mandatory Microstructure Transparency:** Platforms should be required to publish real-time bid-ask depth, wallet concentration metrics, and automated market maker fee distributions to allow journalists to distinguish between retail consensus and individual whale bets.
- **Media Sourcing Standards:** Financial journalism codes should mandate reporting order book depth and spreads alongside nominal probabilities, preventing single large block trades from being misreported as broad democratic shifts.
- **Circuit Breakers on Event Contracts:** Introducing dynamic price bands during presidential debates and vote-tallying windows can prevent flash crashes from generating spurious national news narratives.

9 Conclusion

This paper provides the first micro-founded theoretical framework and structural causal evidence quantifying the reflexivity loop between decentralized prediction markets and media coverage. By discarding flawed whale trade instruments in favor of [Rigobon \(2003\)](#) Identification through Heteroskedasticity and integrating high-frequency Polymarket CLOB, FiveThirtyEight polling, and Wikimedia attention data across seven 2024 battleground states, we establish that prediction markets are far more than passive mirrors of public sentiment.

A 10 percentage point surge in odds generates a +54.8% explosion in media attention, which subsequently feeds back into market pricing and voter coordination. These findings bridge market microstructure, behavioral media economics, and political economy, providing rigorous empirical foundations for the future governance of digital financial markets and democratic elections.

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